

# Programming Assignment 3

## R Programming

### Introduction

Download the file `ProgAssignment3-data.zip` file containing the data for Programming Assignment 3 from the Coursera web site. Unzip the file in a directory that will serve as your working directory. When you start up R make sure to change your working directory to the directory where you unzipped the data.

The data for this assignment come from the Hospital Compare web site (<http://hospitalcompare.hhs.gov>) run by the U.S. Department of Health and Human Services. The purpose of the web site is to provide data and information about the quality of care at over 4,000 Medicare-certified hospitals in the U.S. This dataset essentially covers all major U.S. hospitals. This dataset is used for a variety of purposes, including determining whether hospitals should be fined for not providing high quality care to patients (see <http://goo.gl/jAXFX> for some background on this particular topic).

The Hospital Compare web site contains a lot of data and we will only look at a small subset for this assignment. The zip file for this assignment contains three files

- `outcome-of-care-measures.csv`: Contains information about 30-day mortality and readmission rates for heart attacks, heart failure, and pneumonia for over 4,000 hospitals.
- `hospital-data.csv`: Contains information about each hospital.
- `Hospital_Revised_Flatfiles.pdf`: Descriptions of the variables in each file (i.e the code book).

A description of the variables in each of the files is in the included PDF file named `Hospital_Revised_Flatfiles.pdf`. This document contains information about many other files that are not included with this programming assignment. You will want to focus on the variables for Number 19 (“Outcome of Care Measures.csv”) and Number 11 (“Hospital\_Data.csv”). You may find it useful to print out this document (at least the pages for Tables 19 and 11) to have next to you while you work on this assignment. In particular, the numbers of the variables for each table indicate column indices in each table (i.e. “Hospital Name” is column 2 in the `outcome-of-care-measures.csv` file).

## 1 Plot the 30-day mortality rates for heart attack

Read the outcome data into R via the `read.csv` function and look at the first few rows.

```
> outcome <- read.csv("outcome-of-care-measures.csv", colClasses = "character")
> head(outcome)
```

There are many columns in this dataset. You can see how many by typing `ncol(outcome)` (you can see the number of rows with the `nrow` function). In addition, you can see the names of each column by typing `names(outcome)` (the names are also in the PDF document).

To make a simple histogram of the 30-day death rates from heart attack (column 11 in the outcome dataset), run

```
> outcome[, 11] <- as.numeric(outcome[, 11])
> ## You may get a warning about NAs being introduced; that is okay
> hist(outcome[, 11])
```

Because we originally read the data in as character (by specifying `colClasses = "character"` we need to coerce the column to be numeric. You may get a warning about NAs being introduced but that is okay.

## 2 Finding the best hospital in a state

Write a function called `best` that take two arguments: the 2-character abbreviated name of a state and an outcome name. The function reads the `outcome-of-care-measures.csv` file and returns a character vector with the name of the hospital that has the best (i.e. lowest) 30-day mortality for the specified outcome in that state. The hospital name is the name provided in the `Hospital.Name` variable. The outcomes can be one of “heart attack”, “heart failure”, or “pneumonia”. Hospitals that do not have data on a particular outcome should be excluded from the set of hospitals when deciding the rankings.

**Handling ties.** If there is a tie for the best hospital for a given outcome, then the hospital names should be sorted in alphabetical order and the first hospital in that set should be chosen (i.e. if hospitals “b”, “c”, and “f” are tied for best, then hospital “b” should be returned).

The function should use the following template.

```
best <- function(state, outcome) {  
  ## Read outcome data  
  
  ## Check that state and outcome are valid  
  
  ## Return hospital name in that state with lowest 30-day death  
  ## rate  
}
```

The function should check the validity of its arguments. If an invalid `state` value is passed to `best`, the function should throw an error via the `stop` function with the exact message “invalid state”. If an invalid `outcome` value is passed to `best`, the function should throw an error via the `stop` function with the exact message “invalid outcome”.

Here is some sample output from the function.

```
> source("best.R")  
> best("TX", "heart attack")  
[1] "CYPRESS FAIRBANKS MEDICAL CENTER"  
> best("TX", "heart failure")  
[1] "FORT DUNCAN MEDICAL CENTER"  
> best("MD", "heart attack")  
[1] "JOHNS HOPKINS HOSPITAL, THE"  
> best("MD", "pneumonia")  
[1] "GREATER BALTIMORE MEDICAL CENTER"  
> best("BB", "heart attack")  
Error in best("BB", "heart attack") : invalid state  
> best("NY", "hert attack")  
Error in best("NY", "hert attack") : invalid outcome  
>
```

Save your code for this function to a file named `best.R`.

### 3 Ranking hospitals by outcome in a state

Write a function called `rankhospital` that takes three arguments: the 2-character abbreviated name of a state (`state`), an outcome (`outcome`), and the ranking of a hospital in that state for that outcome (`num`). The function reads the `outcome-of-care-measures.csv` file and returns a character vector with the name of the hospital that has the ranking specified by the `num` argument. For example, the call

```
rankhospital("MD", "heart failure", 5)
```

would return a character vector containing the name of the hospital with the 5th lowest 30-day death rate for heart failure. The `num` argument can take values “best”, “worst”, or an integer indicating the ranking (smaller numbers are better). If the number given by `num` is larger than the number of hospitals in that state, then the function should return `NA`. Hospitals that do not have data on a particular outcome should be excluded from the set of hospitals when deciding the rankings.

**Handling ties.** It may occur that multiple hospitals have the same 30-day mortality rate for a given cause of death. In those cases ties should be broken by using the hospital name. For example, in Texas (“TX”), the hospitals with lowest 30-day mortality rate for heart failure are shown here.

```
> head(texas)
```

|      | Hospital.Name                    | Rate | Rank |
|------|----------------------------------|------|------|
| 3935 | FORT DUNCAN MEDICAL CENTER       | 8.1  | 1    |
| 4085 | TOMBALL REGIONAL MEDICAL CENTER  | 8.5  | 2    |
| 4103 | CYPRESS FAIRBANKS MEDICAL CENTER | 8.7  | 3    |
| 3954 | DETAR HOSPITAL NAVARRO           | 8.7  | 4    |
| 4010 | METHODIST HOSPITAL,THE           | 8.8  | 5    |
| 3962 | MISSION REGIONAL MEDICAL CENTER  | 8.8  | 6    |

Note that Cypress Fairbanks Medical Center and Detar Hospital Navarro both have the same 30-day rate (8.7). However, because Cypress comes before Detar alphabetically, Cypress is ranked number 3 in this scheme and Detar is ranked number 4. One can use the `order` function to sort multiple vectors in this manner (i.e. where one vector is used to break ties in another vector).

The function should use the following template.

```
rankhospital <- function(state, outcome, num = "best") {  
  ## Read outcome data  
  
  ## Check that state and outcome are valid  
  
  ## Return hospital name in that state with the given rank  
  ## 30-day death rate  
}
```

The function should check the validity of its arguments. If an invalid `state` value is passed to `rankhospital`, the function should throw an error via the `stop` function with the exact message “invalid state”. If an invalid `outcome` value is passed to `rankhospital`, the function should throw an error via the `stop` function with the exact message “invalid outcome”.

Here is some sample output from the function.

```
> source("rankhospital.R")  
> rankhospital("TX", "heart failure", 4)  
[1] "DETAR HOSPITAL NAVARRO"  
  
> rankhospital("MD", "heart attack", "worst")
```

```
[1] "HARFORD MEMORIAL HOSPITAL"

> rankhospital("MN", "heart attack", 5000)

[1] NA
```

Save your code for this function to a file named `rankhospital.R`.

## 4 Ranking hospitals in all states

Write a function called `rankall` that takes two arguments: an outcome name (`outcome`) and a hospital ranking (`num`). The function reads the `outcome-of-care-measures.csv` file and returns a 2-column data frame containing the hospital in each state that has the ranking specified in `num`. For example the function call `rankall("heart attack", "best")` would return a data frame containing the names of the hospitals that are the best in their respective states for 30-day heart attack death rates. The function should return a value for every state (some may be `NA`). The first column in the data frame is named `hospital`, which contains the hospital name, and the second column is named `state`, which contains the 2-character abbreviation for the state name. Hospitals that do not have data on a particular outcome should be excluded from the set of hospitals when deciding the rankings.

**Handling ties.** The `rankall` function should handle ties in the 30-day mortality rates in the same way that the `rankhospital` function handles ties.

The function should use the following template.

```
rankall <- function(outcome, num = "best") {
  ## Read outcome data

  ## Check that state and outcome are valid

  ## For each state, find the hospital of the given rank

  ## Return a data frame with the hospital names and the
  ## (abbreviated) state name
}
```

**NOTE:** For the purpose of this part of the assignment (and for efficiency), your function should NOT call the `rankhospital` function from the previous section.

The function should check the validity of its arguments. If an invalid `outcome` value is passed to `rankall`, the function should throw an error via the `stop` function with the exact message “invalid outcome”. The `num` variable can take values “best”, “worst”, or an integer indicating the ranking (smaller numbers are better). If the number given by `num` is larger than the number of hospitals in that state, then the function should return `NA`.

Here is some sample output from the function.

```
> source("rankall.R")
> head(rankall("heart attack", 20), 10)

      hospital state
AK          <NA>   AK
AL      D W MCMILLAN MEMORIAL HOSPITAL   AL
AR  ARKANSAS METHODIST MEDICAL CENTER   AR
```

|    |                                     |    |
|----|-------------------------------------|----|
| AZ | JOHN C LINCOLN DEER VALLEY HOSPITAL | AZ |
| CA | SHERMAN OAKS HOSPITAL               | CA |
| CO | SKY RIDGE MEDICAL CENTER            | CO |
| CT | MIDSTATE MEDICAL CENTER             | CT |
| DC | <NA>                                | DC |
| DE | <NA>                                | DE |
| FL | SOUTH FLORIDA BAPTIST HOSPITAL      | FL |

```
> tail(rankall("pneumonia", "worst"), 3)
```

|    | hospital                                   | state |
|----|--------------------------------------------|-------|
| WI | MAYO CLINIC HEALTH SYSTEM - NORTHLAND, INC | WI    |
| WV | PLATEAU MEDICAL CENTER                     | WV    |
| WY | NORTH BIG HORN HOSPITAL DISTRICT           | WY    |

```
> tail(rankall("heart failure"), 10)
```

|    | hospital                                                          | state |
|----|-------------------------------------------------------------------|-------|
| TN | WELLMONT HAWKINS COUNTY MEMORIAL HOSPITAL                         | TN    |
| TX | FORT DUNCAN MEDICAL CENTER                                        | TX    |
| UT | VA SALT LAKE CITY HEALTHCARE - GEORGE E. WAHLEN VA MEDICAL CENTER | UT    |
| VA | SENTARA POTOMAC HOSPITAL                                          | VA    |
| VI | GOV JUAN F LUIS HOSPITAL & MEDICAL CTR                            | VI    |
| VT | SPRINGFIELD HOSPITAL                                              | VT    |
| WA | HARBORVIEW MEDICAL CENTER                                         | WA    |
| WI | AURORA ST LUKES MEDICAL CENTER                                    | WI    |
| WV | FAIRMONT GENERAL HOSPITAL                                         | WV    |
| WY | CHEYENNE VA MEDICAL CENTER                                        | WY    |

Save your code for this function to a file named `rankall.R`.